

Performance Testing

Date	
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	HDI Prediction API (/predict)
Test Environment	Local (Flask Application)
Test Date	July 2 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits HDI prediction request	1	30	Prediction generated successfully
2	Multiple users access prediction API simultaneously	10	60	All requests processed without errors
3	Continuous prediction requests	20	120	Stable response with acceptable response time
4	Invalid input values	5	30	Validation error message displayed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.1 sec	Pass	Within target
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Stable performance
3	Throughput (Req/sec)	> 15 Req/sec	18 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	42%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	58%	Pass	Memory usage within limits

Step 4: Observations & Analysis

Key Findings

- The HDI prediction API responded quickly under normal and moderate load.
- No request failures occurred during testing.
- The application remained stable throughout the test.

Bottlenecks Identified

- Response time increased slightly as concurrent requests increased.
- Performance depends on local system resources.

Optimization Steps Taken

- Optimized input validation.
- Improved model loading efficiency.
- Reduced unnecessary processing before prediction.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Human Development Index - Prediction

Human Development Index

Select the name of the Country

Enter life expectancy rate (range 40-80)

Enter Mean years of schooling (range 4-16)

Enter Expected years of schooling (range 0-21)

Enter GNI per capita (range 100-150000)

Predict

Very High HDI 0.84

Human Development Index

Human Development Index

The Human Development Index (HDI) is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable, and having a decent standard of living.

It is calculated using indicators such as life expectancy, mean years of schooling, expected years of schooling, and Gross National Income (GNI) per capita — combining these into a single score between 0 and 1.

This application uses a trained Machine Learning model (Linear Regression) to predict the HDI score of a country based on its key development indicators. Click **Predict** above to try it yourself.